ReactorGenome

One Digital Twin for the Batch, CSTR, PFR and RTD Labs

CHE331 Innovation Project: Final Report

Mayur Dilip Pachpute
Roll No. 240643

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Summary. ReactorGenome is a web application that sits beside the CHE331 reactor rigs. Students type the readings they already take; the app fits them to the reactor equations, shows what each textbook assumption costs, and lets the fitted kinetics drive reactor networks that cannot be built in the lab. All five parts of the proposal were built. Three points are only partly met and are listed openly in Section 3. On the 2026 lab reports the app reproduces every reported rate constant and activation energy to four significant figures, and 134 automated tests check the engine, the pages and the exports.

Source code: `https://github.com/Mayurdp2403/ReactorGenome`

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1 The problem

All four reactor experiments in CHE331 (Batch, CSTR, PFR and RTD) are run at fixed conditions: one reactor, three flow rates, three temperatures and equal dilute feeds of NaOH and ethyl acetate. Readings are written down and analysed later, so nobody sees what is happening inside the reactor while the rig is running. Questions such as why conversion falls with flow, what a dead zone does, or why one residence time is not enough to reach steady state are answered in words, never tested. In our own CSTR run the conductivity was still drifting when the "steady" readings were taken, and 28 °C gave more conversion than 40 °C. We had no way to find out why.

The proposal set out to build one web app, used on a laptop next to the rig with no new hardware, that turns the manual readings into proper fits, live plots, assumption checks and what-if studies.

2 What was built

ReactorGenome is a Python application (Streamlit front end, NumPy/SciPy engine) with one page per experiment and a Design Studio for reactor networks. It runs on a laptop with `run.bat`, and other laptops on the same network can open the same address.

Every experiment page follows the same layout (Figure 1): assumptions on the left, an animated reactor in the middle, results and diagnostics on the right, the readings table below, then the analysis tabs. Every control has a short help text.

The code has two layers:

- **Engine** (`reactorgenome/engine`): the models and solvers, with no user-interface code. It covers units and bounds, rate laws, the stirred tank (steady, transient, non-ideal mixing and energy balance), batch fitting, plug flow with dispersion and laminar flow, RTD moments and model fits, the lab-sheet analysis and a flowsheet solver.
- **Interface** (`reactorgenome/ui`): one module per page, plus shared widgets, charts, cached engine calls and exports. The animation player and the drag-and-drop design board are small HTML/JavaScript components.

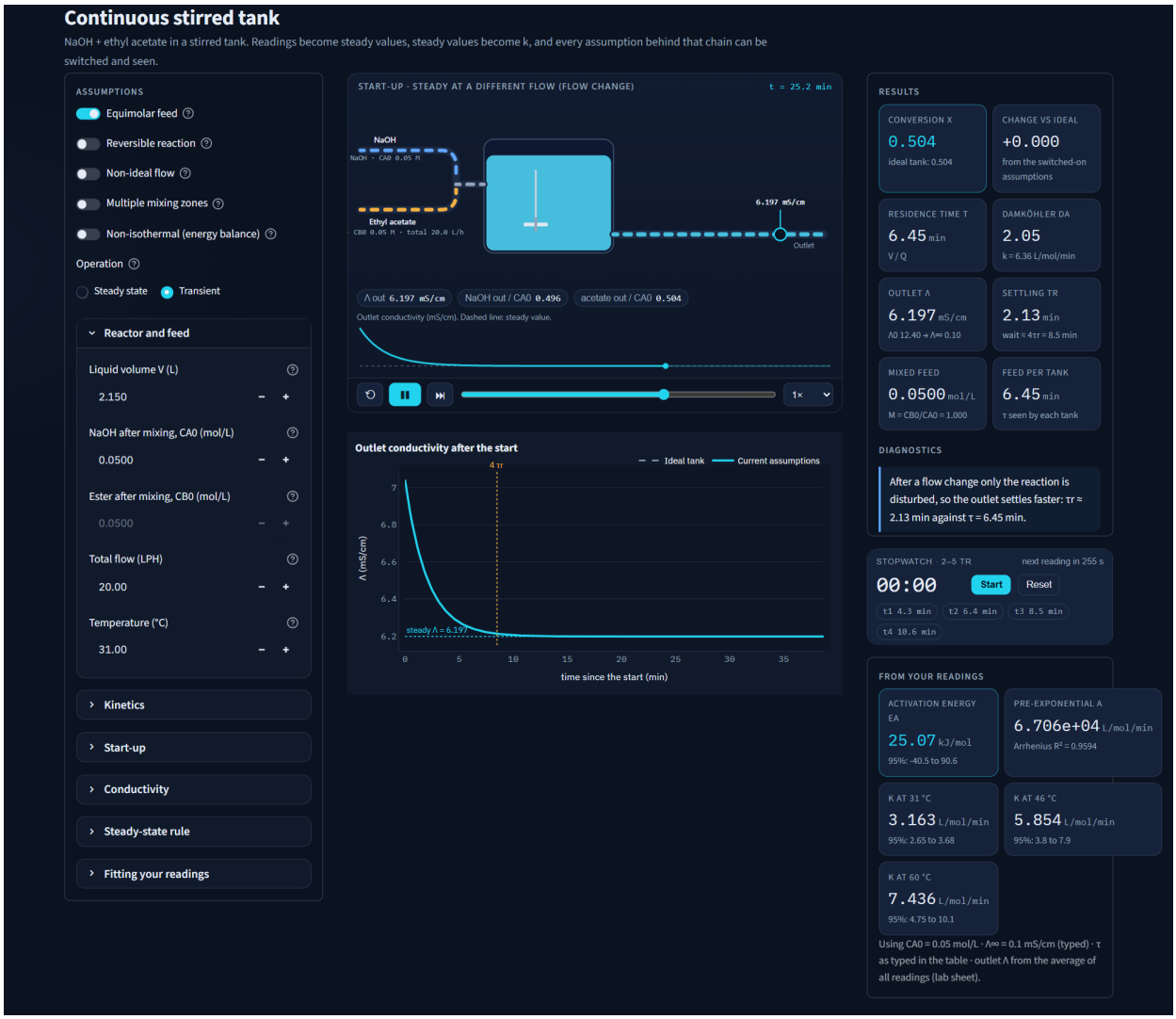


Figure 1: CSTR page. Left: assumption switches and run conditions. Middle: animated tank with the simulated start-up. Right: results, diagnostics and the kinetics fitted from the readings typed below (Ea = 25.07 kJ/mol).

3 Proposal against delivery

Table 1 takes each promise in the proposal in turn. "Partly" means the feature exists but does not yet do everything the proposal describes.

Table 1: What the proposal promised and what was delivered.

Proposal	Delivered	Status
1. Enter the readings		
Pick the rig and type readings at the times a stopwatch gives	Batch, CSTR, PFR and RTD pages. The readings tables follow the lab sheets and accept CSV import. A stopwatch on the CSTR page gives the reading schedule (2–5 settling times).	**Delivered**
2. Fit the data properly		
Treat CSTR readings as a transient and extrapolate to the true steady state	Plateau from the simulated start-up shape, a single exponential, or the manual average. When reading times are typed, each run gets a steady or not-steady verdict.	**Delivered**

Proposal	Delivered	Status
Conductivity to conversion from ion concentrations, not a fixed Λ_∞	Default $\Lambda_\infty = 0.366\,\Lambda_0$ from limiting ionic conductivities. A typed value is also accepted, and Λ_∞ can be estimated from the readings with a 95% interval.	**Delivered**
Fit dead volume, bypass and mixing zones	The bypass fraction is fitted from the readings. Dead volume, stagnant zones and tanks in series are set by the user and compared against the readings. The RTD page measures N and Pe directly.	**Partly**
An error bar on every number	95% intervals on k, E_a and A, error bars on measured conversions, and parity and residual plots.	**Delivered**
3. Watch it live		
Design plot, Arrhenius plot and transient redraw on each reading	All plots and the fitted kinetics update as each cell is typed.	**Delivered**
Show the expected reading first, then judge ideal or not	The app shows the outlet conductivity and conversion it expects for the current settings. Diagnostics after the fit explain non-ideal behaviour. There is no step-by-step prediction before each individual reading.	**Partly**
4. Change the assumptions		
Switches for equimolar feed, irreversibility, ideal mixing, isothermal operation and steady state after one τ	Each assumption is a switch on its page. The *Assumption effects* tab shows the conversion with each one dropped on its own (Figure 5).	**Delivered**
5. Try what the lab cannot do		
Reactors in series in either order, unequal sizes, recycle, non-equimolar feeds, packed bed as a reactor	Design Studio: a drag-and-connect board solved as one network, with presets and a mass-balance check.	**Delivered**
Non-isothermal scale-up with runaway and multiple steady states; Levenspiel plots and phase portraits	Energy-balance tab with the heat-generation and heat-removal curves, stability of each steady state and phase portraits. Levenspiel plots are on the PFR page and in the Design Studio.	**Delivered**
Batch k improves the CSTR and PFR predictions in the same session	Fitted second-order kinetics are published to every page and to the Design Studio.	**Delivered**
All four experiments on one conversion–Da plot	The CSTR page plots its measured runs on the X–Da curve. The Design Studio overlays $k(T)$ from every rig and shows the ideal Da curves, but without the measured points from each rig.	**Partly**
Deliverables		
Four-tab app usable on a laptop or phone within the lab slot	Four experiment pages plus the Design Studio; runs locally and over the lab network. Laptop use was tested; phone layout was not.	**Delivered**
One-click export of the observation table, results, graphs and a diagnosis	Each page exports a ZIP: an offline HTML report, CSV tables, every plot and every setting. The whole session saves as a project file.	**Delivered**

4 The application, page by page

The screenshots below are the app running on the readings from the 2026 CHE331 lab reports. Those readings were used only to check the mathematics. They are not built into the app, which starts with empty tables.

4.1 CSTR

Readings are typed exactly as on the lab sheet (Figure 2): one row per run with temperature, total flow, τ and up to four outlet conductivities, plus a small table of Λ_0 for each temperature.

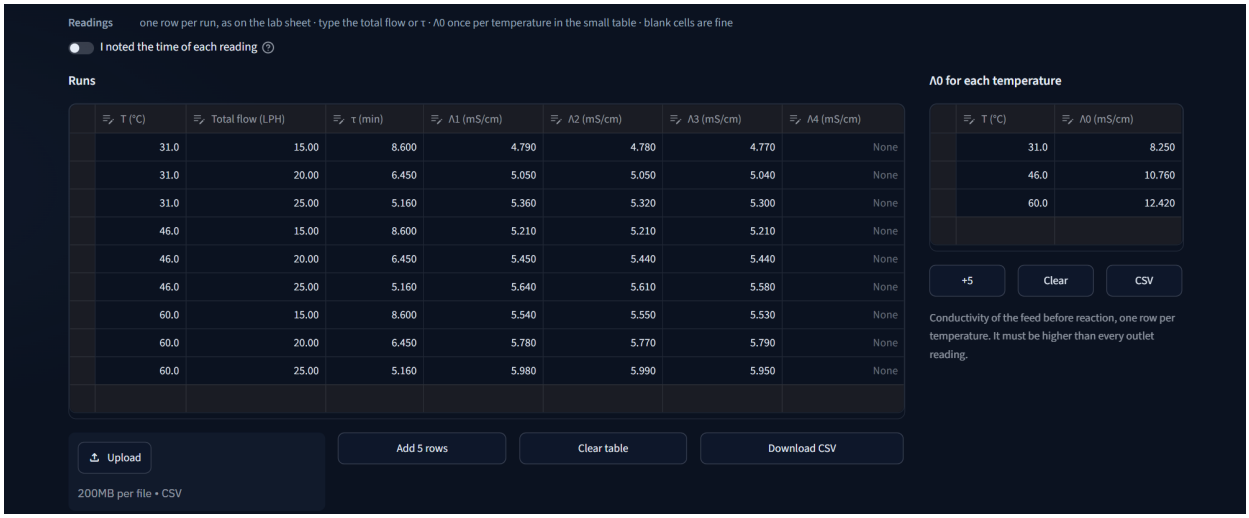


Figure 2: Readings table, laid out like the lab sheet's observation table.

The *Results* tab (Figure 3) collects everything the lab report needs on one screen: E_a, A and k at each temperature with 95% intervals, the design plot for each temperature and for all of them together, the Arrhenius plot, conversion against flow and temperature, single-run k against τ, a parity plot and residuals. The rate law used for the fit (order 1 to 3) is a setting.

Below the plots, the fitted $k(T)$ predicts the conversion at any temperature, feed concentration and τ (Figure 4). The right-hand plot shows why one concentration cannot pin down the reaction order: every order fits the runs at the C_{A0} that was used, and the orders separate only away from it.

The energy balance turns the tank into the scale-up case from the proposal. With a concentrated feed and a weak jacket (Figure 6) the heat-generation curve crosses the heat-removal line three times. The steady states at 31 °C and 134 °C are stable; the one at 80 °C is not. The phase portrait shows which start-up conditions run away to the hot state.



Figure 3: Results tab: kinetics with intervals and all of the report's plots on one screen.

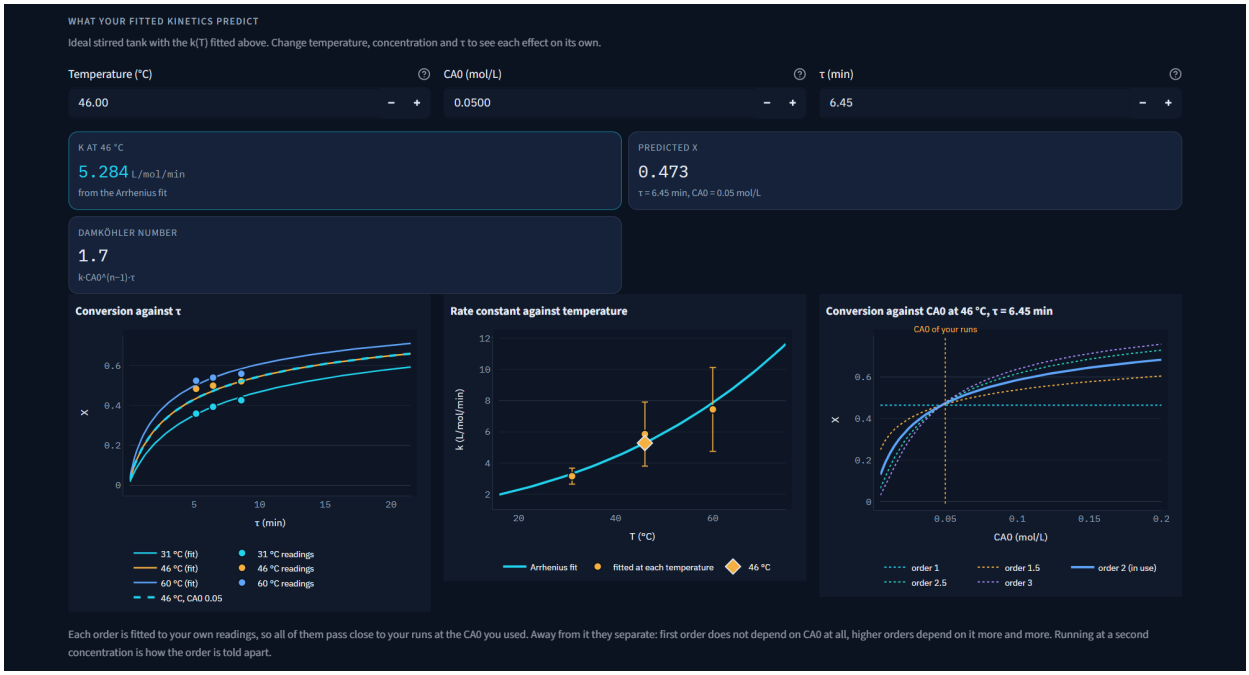


Figure 4: What the fitted kinetics predict: X against τ, k against T, and X against C_{A0} for orders 1 to 3.

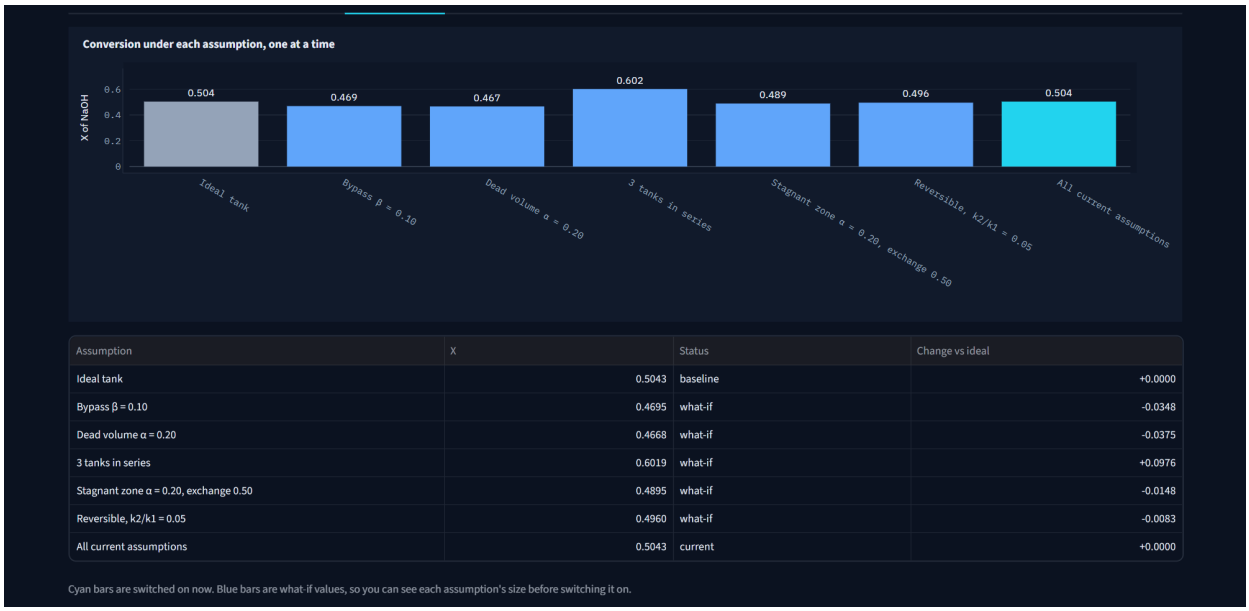


Figure 5: Assumption effects: conversion with each textbook assumption dropped on its own.

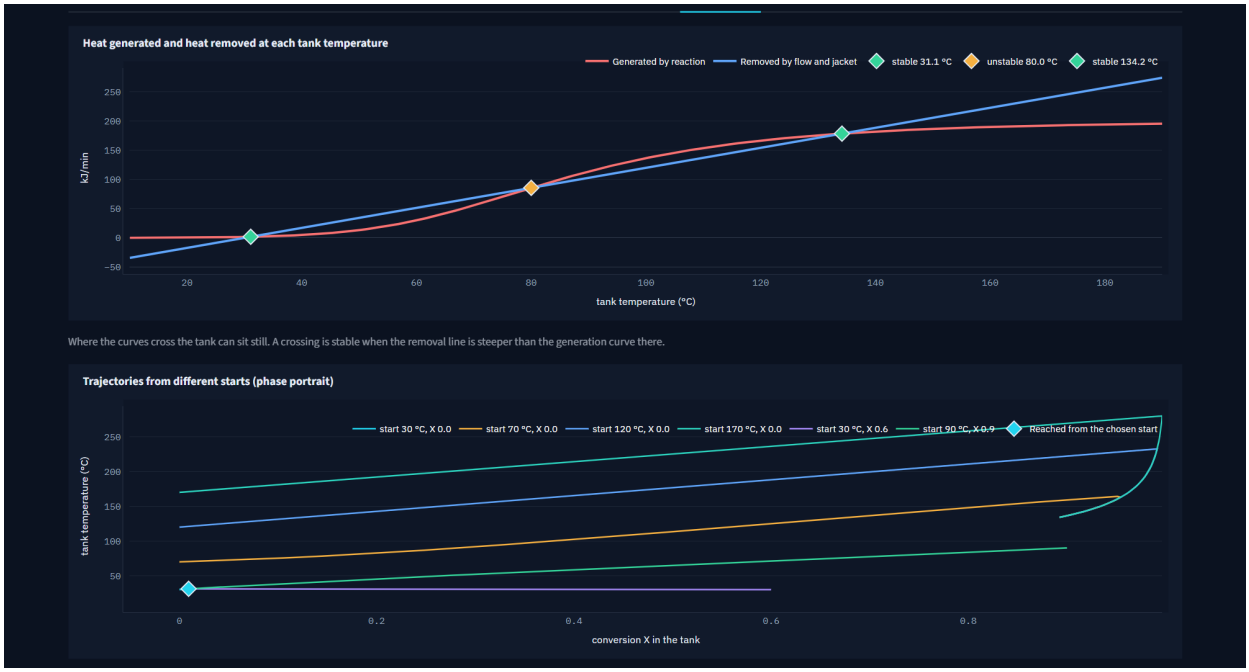


Figure 6: Energy balance for a 5 mol/L feed with a weak jacket ($UA = 40$ J/min/K, $E_a = 90$ kJ/mol, $\Delta H = -100$ kJ/mol): three steady states and the phase portrait.

4.2 PFR

The PFR page treats the helical coil as a plug-flow reactor and lets the user switch on axial dispersion, laminar flow, recycle, an adiabatic temperature rise and reaction that continues in the sample beaker. Coil Reynolds and Dean numbers are worked out from the coil geometry.

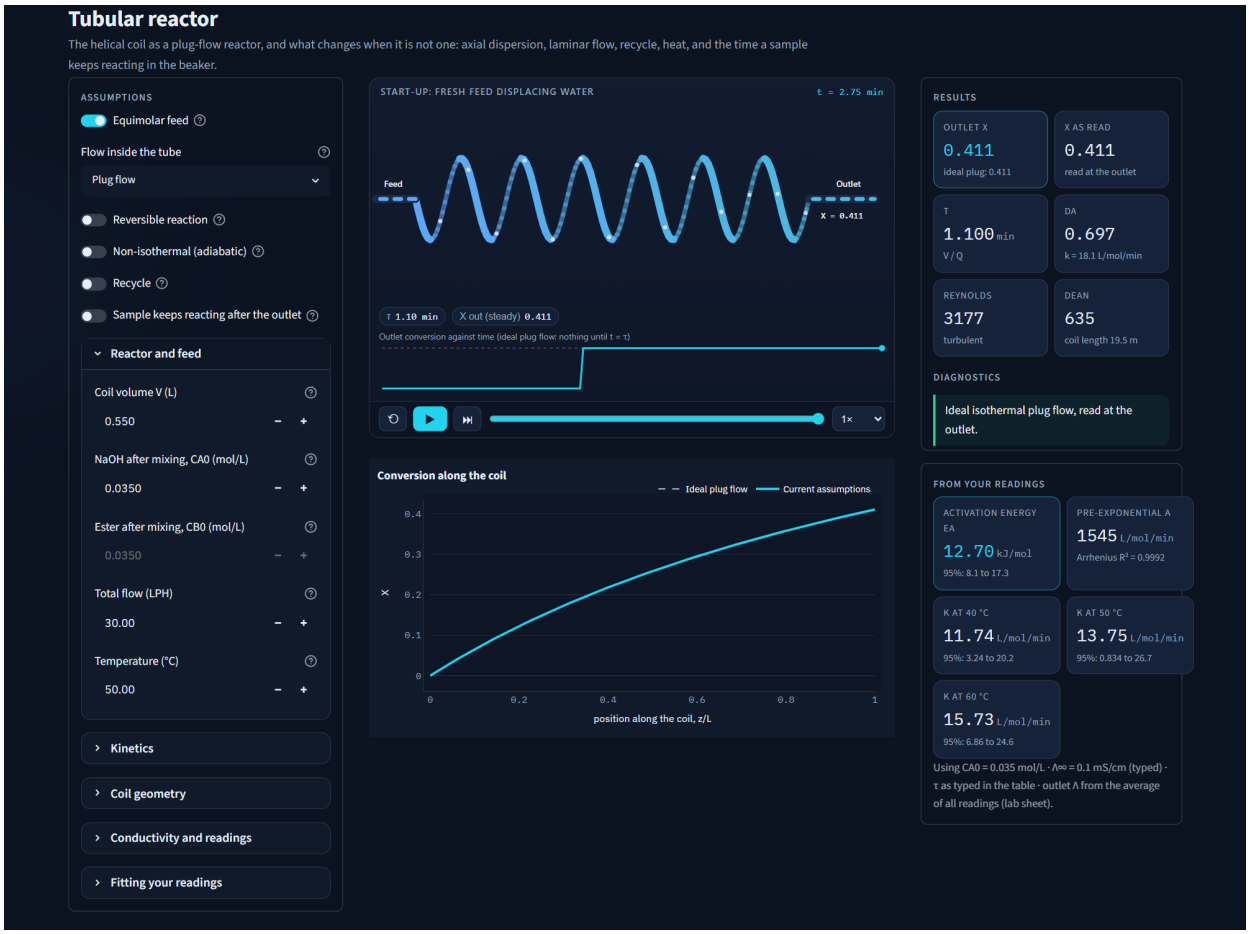


Figure 7: PFR page with the kinetics fitted from the readings ($E_a = 12.70$ kJ/mol, as in the lab report).



Figure 8: PFR results: design plots, Arrhenius plot, conversion against flow and temperature, single-run k, parity and residuals.

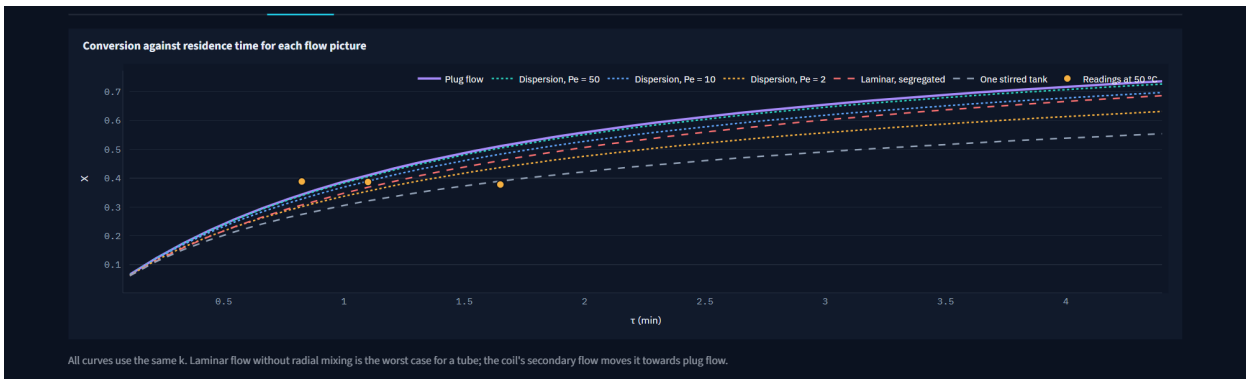


Figure 9: The same k under each flow picture, with the measured points at 50 °C.

4.3 Batch

The batch page fits conductivity against time. The rate law is a choice, not an assumption: first order, second order, reversible and a fitted power law are compared by AIC, with a clock offset for the seconds between mixing and starting the stopwatch.



Figure 10: Batch page at 50.2 °C: fitted $k = 102.9$ L/mol/min with its interval.



Figure 11: Batch fit: integral and differential plots, residuals, and the rate laws ranked by AIC. The differential method gives $n \approx 1.0$, the same as the lab report.

4.4 RTD

Up to four tracer runs share one time column. For each run the page gives the mean time, the variance, N (both the exact value and the nearest whole number), the Peclet number under three conventions, and fitted tanks-in-series and dispersion models. It also warns when the curve was cut off before the tracer had left.

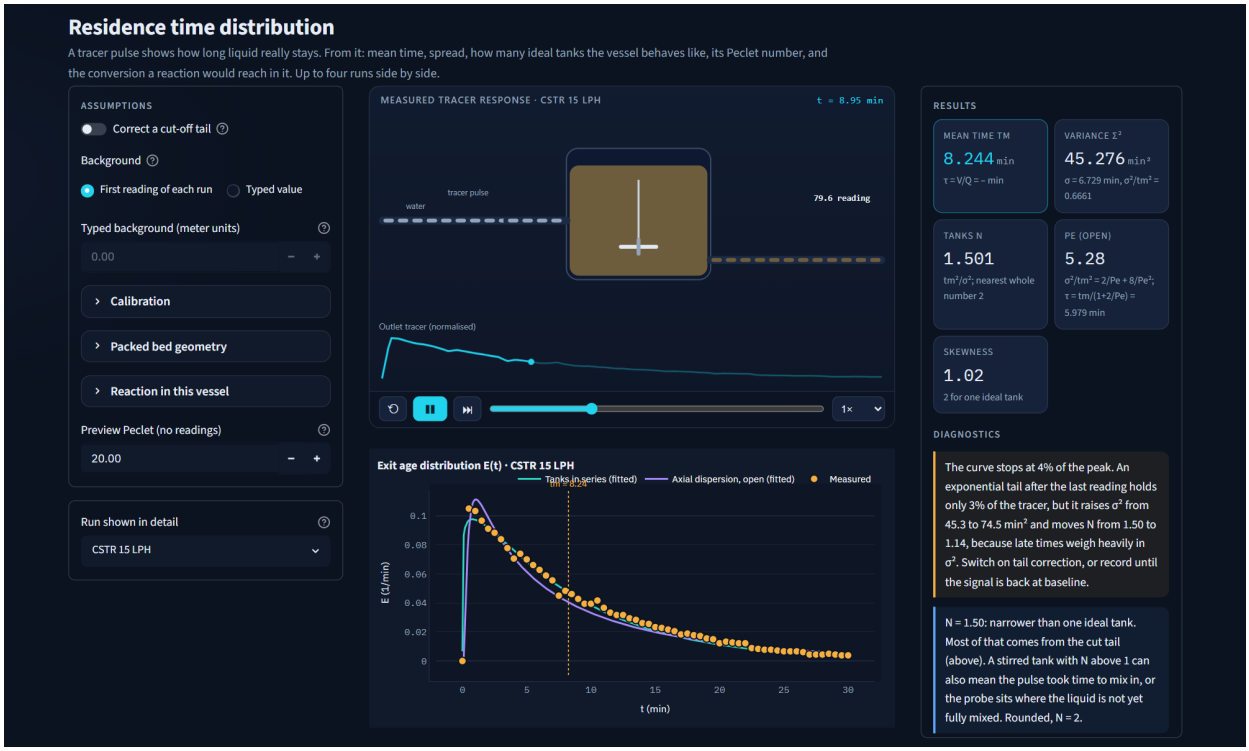


Figure 12: RTD page, stirred tank at 15 LPH. The diagnostic explains that $N = 1.5$ comes mostly from the cut-off tail.



Figure 13: All four runs: tracer curves, $E(t)$ for each vessel, $E(\theta)$ and $F(t)$.



Figure 14: Model fits ranked by error, with the residuals of the best model.

4.5 Design Studio

Blocks (feeds, CSTRs, PFRs, packed beds, splitters, products) are dragged onto a board, connected and solved as one network. Recycle loops converge by Wegstein acceleration, and the mass balance closes to better than 10^{-8}. The same kinetics can come from any lab page.

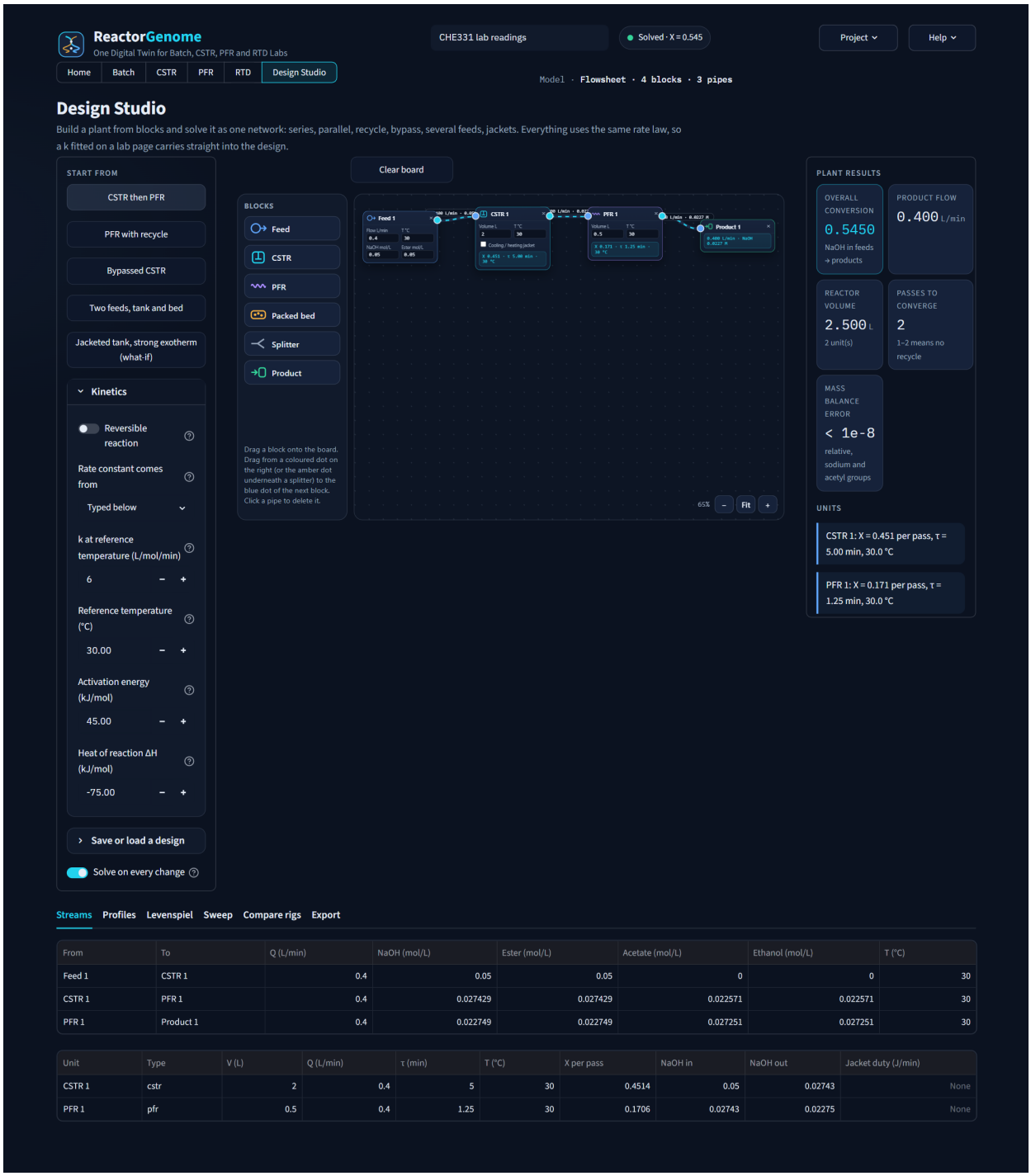


Figure 15: Design Studio with a CSTR followed by a PFR: overall conversion 0.545, stream table and unit table.

4.6 Export

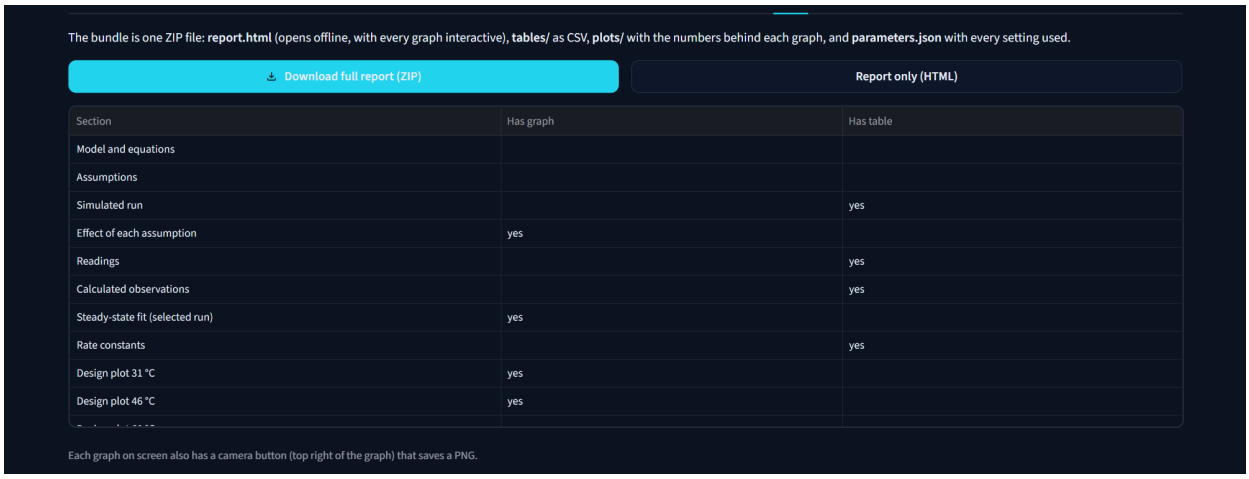


Figure 16: Export tab: one ZIP with an offline HTML report, CSV tables, every plot and every setting used.

5 Validation

The engine was checked in two ways: against closed-form results, and against the numbers in the 2026 lab reports analysed with each report's own conventions. The full record, which a script regenerates from fresh calculations, is in `docs/VALIDATION.md`.

Table 2: Engine against closed-form results (selected).

Case	Engine	Exact	Error
Ideal CSTR, $Da = 1$	0.3820	0.3820	1×10^{-16}
Ideal CSTR, $Da = 1$, $M = 2$	0.5858	0.5858	4×10^{-14}
Plug flow, $X = Da/(1 + Da)$	0.2308	0.2308	8×10^{-12}
Dispersion, $Pe = 1000$ (plug-flow limit)	0.2307	0.2308	9×10^{-5}
Transient CSTR settles on the algebraic steady state	0.4637	0.4637	5×10^{-14}
Flowsheet CSTR $\rightarrow$ PFR against sequential solution	0.5450	0.5450	5×10^{-10}
RTD of 3 tanks: N from moments	3.000	3	–

Table 3: App against the 2026 lab reports, using each report's own conventions.

Experiment	Quantity	Report	App
CSTR	k at 31 / 46 / 60 °C (L/mol/min)	3.1629 / 5.8541 / 7.4362	3.1629 / 5.8541 / 7.4362
	E_a (kJ/mol), A (L/mol/min)	25.07, 6.71×10^4	25.07, 6.71×10^4
PFR	k at 40 / 50 / 60 °C (L/mol/min)	11.738 / 13.749 / 15.730	11.738 / 13.749 / 15.730
	E_a (kJ/mol)	12.70	12.70
RTD	CSTR 15 LPH: t_m (min), σ^2 (min^2), N	8.244, 45.276, 1.501	8.244, 45.276, 1.501
	Packed bed 15 LPH: Pe, τ (min)	35.93, 4.896	35.93, 4.896
Batch	Reaction order (differential method)	0.987	1.01

The app reproduces every reported value. Applying the app's own methods to the same readings showed things the reports could not:

- **Λ_∞ is the largest source of error.** Fitting one Λ_∞ to all nine CSTR runs gives $\Lambda_\infty \approx 0.30\,\Lambda_0$, close to the ion estimate of $0.366\,\Lambda_0$ and far from the 0.1 mS/cm used in the report. With it, k at 31 °C is about 8.5 L/mol/min rather than 3.16, and the single-run k no longer drifts with

τ.

- **CSTR steady state.** Extrapolating each run to its plateau removes most of the "28 °C beats 40 °C" inversion described in the proposal. The inversion came from averaging readings taken before the tank had settled, not from the kinetics.
- **RTD tail.** The stirred-tank curves stop at 4% of the peak. Adding the missing tail brings N from 1.5 to about 1.2, close to one ideal tank.
- **Batch order.** After the mixing period the batch trace fits first order better than second order, which agrees with the report's own differential method.

6 Beyond the proposal

- **Reaction order as a setting** (1 to 3) on the CSTR and PFR pages, with a table comparing how well each order fits and a plot showing where the orders separate.
- **Three ways to fit** k (conversion function on τ, τ on conversion function, a free line), each with intervals, parity plots and residuals.
- **Single-run** k **against** τ as a consistency check: a sloping line points to Λ_∞, a sampling delay, or runs that were not steady.
- **RTD extras:** tail correction with the tail's share of the tracer, three Peclet conventions, τ from Pe, D_L from the bed geometry, and the conversion each measured vessel would give.
- **Sample delay in the PFR,** modelled explicitly instead of hidden in a free intercept of 7 to 16 minutes.
- **Input checks:** each page states the inputs it used and flags rows that cannot be analysed, such as a missing Λ_0 or readings beyond Λ_∞.
- **Project files:** the whole session (tables, settings and the design board) saves to one file and loads back.

7 Testing

The test suite (`python -m pytest tests`, 134 tests) covers:

- closed-form limits for every reactor model, unit conversions and bounds, invalid inputs and solver failures;
- flowsheets with recycle, the RTD moments and model fits;
- each lab report's numbers, reproduced to four significant figures;
- every page rendered with and without readings, including malformed tables;
- the exports, and saving and restoring a project.

`tools/make_validation.py` regenerates the validation record from fresh calculations.

8 Limitations

- The three "Partly" items in Table 1: dead volume and stagnant zones are compared rather than fitted, there is no prediction step before each individual reading, and the common Da plot does not yet carry the measured points from every rig.

- The layout has not been tested on a phone.
- Only second-order kinetics are shared between pages. Fits with another order stay on the page that made them.
- The energy balance is checked against its isothermal limit and for multiple steady states, not against measured temperatures, because the rigs are run isothermally.
- The chemistry is NaOH with ethyl acetate throughout, as in the course.

9 Running the app

```
git clone https://github.com/Mayurdp2403/ReactorGenome.git
cd ReactorGenome
python -m pip install -r requirements.txt
python -m streamlit run app.py
```

On Windows, `run.bat` does the same. The screenshots in this report are regenerated with `python docs/report/make_figures.py`.